

2. In the term *trace element*, the adjective *trace* means that
- (A) the element is required in very small amounts.
 - (B) the element can be used as a label to trace atoms through an organism's metabolism.
 - (C) the element is very rare on Earth.
 - (D) the element enhances health but is not essential for the organism's long-term survival.
3. The reactivity of an atom arises from
- (A) the average distance of the outermost electron shell from the nucleus.
 - (B) the existence of unpaired electrons in the valence shell.
 - (C) the sum of the potential energies of all the electron shells.
 - (D) the potential energy of the valence shell.
4. Which statement is true of all atoms that are anions?
- (A) The atom has more electrons than protons.
 - (B) The atom has more protons than electrons.
 - (C) The atom has fewer protons than does a neutral atom of the same element.
 - (D) The atom has more neutrons than protons.
5. Which of the following statements correctly describes any chemical reaction that has reached equilibrium?
- (A) The concentrations of products and reactants are equal.
 - (B) The reaction is now irreversible.
 - (C) Both forward and reverse reactions have halted.
 - (D) The rates of the forward and reverse reactions are equal.

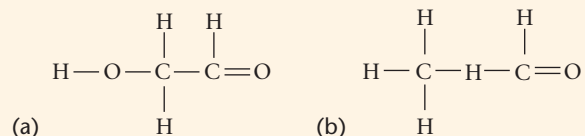
Levels 3-4: Applying/Analyzing

6. We can represent atoms by listing the number of protons, neutrons, and electrons—for example, $2p^+$, $2n^0$, $2e^-$ for helium. Which of the following represents the ^{18}O isotope of oxygen?
- (A) $7p^+$, $2n^0$, $9e^-$
 - (B) $8p^+$, $10n^0$, $8e^-$
 - (C) $9p^+$, $9n^0$, $9e^-$
 - (D) $10p^+$, $8n^0$, $9e^-$
7. The atomic number of sulfur is 16. Sulfur combines with hydrogen by covalent bonding to form a compound, hydrogen sulfide. Based on the number of valence electrons in a sulfur atom, predict the molecular formula of the compound.
- (A) HS
 - (B) HS_2
 - (C) H_2S
 - (D) H_4S
8. What coefficients must be placed in the following blanks so that all atoms are accounted for in the products?



- (A) 2; 1
- (B) 3; 1
- (C) 1; 3
- (D) 2; 2

9. **DRAW IT** Draw Lewis dot structures for each hypothetical molecule shown below, using the correct number of valence electrons for each atom. Determine which molecule makes sense because each atom has a complete valence shell and each bond has the correct number of electrons. Explain what makes the other molecule nonsensical, considering the number of bonds each type of atom can make.



Levels 5-6: Evaluating/Creating

10. **EVOLUTION CONNECTION** The percentages of naturally occurring elements making up the human body (see Table 2.1) are similar to the percentages of these elements found in other organisms. How could you account for this similarity among organisms?

11. **SCIENTIFIC INQUIRY** Female luna moths (*Actias luna*) attract males by emitting chemical signals that spread through the air. A male hundreds of meters away can detect these molecules and fly toward their source. The sensory organs responsible for this behavior are the comblike antennae visible in the photograph shown here. Each filament of an antenna is equipped with thousands of receptor cells that detect the sex attractant. Based on what you learned in this chapter, propose a hypothesis to account for the ability of the male moth to detect a specific molecule in the presence of many other molecules in the air. What predictions does your hypothesis make? Design an experiment to test one of these predictions.



12. **WRITE ABOUT A THEME: ORGANIZATION** While waiting at an airport, Neil Campbell once overheard this claim: "It's paranoid and ignorant to worry about industry or agriculture contaminating the environment with their chemical wastes. After all, this stuff is just made of the same atoms that were already present in our environment." Drawing on your knowledge of electron distribution, bonding, and emergent properties (see Concept 1.1), write a short essay (100–150 words) countering this argument.
13. **SYNTHESIZE YOUR KNOWLEDGE**



This bombardier beetle is spraying a boiling hot liquid that contains irritating chemicals, used as a defense mechanism against its enemies. The beetle stores two sets of chemicals separately in its glands. Using what you learned about chemistry in this chapter, propose a possible explanation for why the beetle is not harmed by the chemicals it stores and what causes the explosive discharge.

For selected answers, see Appendix A.